Direct reciprocity

Four teams, one tournament, and what survived

First class
Five rounds against every other team.

Second class
Name how you played. Rank the teams.

Then
Write each one as a pair (p, q).

Winning team:

The stage game, and the scoreboard

	C	D
C	3	0
D	5	1

1 v 2 (round 1 to 5 | score)

1 ☐☐☐☐☐ ☐

2 ☐☐☐☐☐ ☐

1 v 3

1 ☐☐☐☐☐ ☐

3 ☐☐☐☐☐ ☐

1 v 4

1 ☐☐☐☐☐ ☐

4 ☐☐☐☐☐ ☐

2 v 3

2 ☐☐☐☐☐ ☐

3 ☐☐☐☐☐ ☐

2 v 4

2 ☐☐☐☐☐ ☐

4 ☐☐☐☐☐ ☐

3 v 4

3 ☐☐☐☐☐ ☐

4 ☐☐☐☐☐ ☐

Row player's payoff. $R = 3$, $S = 0$, $T = 5$, $P = 1$. Write each round's C or D as it is played, then the five-round score in the shaded box.

Totals: 1 ☐ 2 ☐ 3 ☐ 4 ☐

While you play

In four teams. You play five rounds against each of the other three teams, and never against yourselves. Both teams choose C or D at the same time each round, and only then see what the other did.

A *reactive* strategy is a pair (p, q): cooperate with probability p after they cooperated, and q after they defected.

Team	How they played	As (p, q)	Rank
1			
2			
3			
4			

$(1, 1)$ is

$(0, 0)$ is

$(1, 0)$ is

$\left(\frac{1}{2}, \frac{1}{2}\right)$ is

Which of the four did the teams actually submit?

Did the winner cooperate first?

Did it ever forgive?

	CC	CD	DC	DD
CC				
CD				
DC				
DD				

Each row has to add to one, which is the check to make as we fill it in.

Rows are where we are now, columns where we go next.

Take $(p, q) = (4/5,\ 1/5)$ against $(p', q') = (3/5,\ 1/10)$.

In state CD I cooperated and they defected, so next round I cooperate with probability [ ] and they cooperate with probability [ ].

The chain settles at $\pi = \frac{1}{245}(26,\ 65,\ 44,\ 110)$. Check one row of $\pi P = \pi$:

My long-run payoff is $\pi \cdot (R, S, T, P) =$ _____, and theirs is _____.

Mutual cooperation for ever would have paid _____ each. What did the reciprocity cost? _____

What we just did in the room is **Question 1** on the Direct Reciprocity page.

(a) Name $(1,1)$, $(0,0)$, $(1,0)$ and $\left(\frac{1}{2},\frac{1}{2}\right)$. [3]

(b) Write the full 4×4 transition matrix for the two strategies. [7]

(c) Confirm the stationary distribution and get each player's long-run payoff. [7]

(d) Why a strategy like Tit For Tat needs care with this method. [8]

vknight.org/gt/topics/direct-reciprocity.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
The tournament stage game	8.2.1	Direct Reciprocity: Definition: Prisoner's Dilemma
A strategy as a pair of probabilities	8.2.3	Direct Reciprocity: Definition: Reactive Strategy
Two strategies meeting repeatedly	8.2.4	Direct Reciprocity: Definition: Markov Chain of Two Reactive Strategies
Scores in the long run	8.2.5	Direct Reciprocity: Theorem: Long-Run Average Payoffs via Stationary Distribution
The steady state we computed	8.2.6	Direct Reciprocity: Theorem: Steady-State Distribution for Two Reactive Strategies

vknight.org/gtb/chapters/direct-reciprocity

Quick quiz before you go: vknight.org/gt/topics/direct-reciprocity.html

Before the next class: read the Direct Reciprocity chapter as far as the steady-state distribution, and attempt Question 1.